

Finite active-set identification can fail

Conjecture 00000008829

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The source asserts finite-time identification of active constraints without specifying an algorithm or imposing nondegeneracy. Under its unqualified universal interpretation, ordinary safe projected gradient is a counterexample. We do not claim that every algorithm must fail to identify the active set. The acceleration and noise clauses are unused.

On the real Hilbert line minimize

$$f(x) = \frac{x^2}{2} \quad \text{subject to} \quad c(x) = -x \leq 0.$$

The feasible set $C = [0, \infty)$ is closed and convex. The objective is smooth, closed and strongly convex with modulus one: its actual gradient is x , and

$$f(y) = f(x) + f'(x)(y - x) + \frac{1}{2}|y - x|^2.$$

The gradient is 1-Lipschitz. The unique constrained minimizer is $x_* = 0$, since $f(x) \geq 0$ with equality only at zero. Its one constraint is active there.

The actual metric projection onto C is $P_C(y) = \max(0, y)$. For $y \geq 0$ this is y itself. For $y < 0$ and every $z \geq 0$, $(y - z)^2 \geq y^2$, so zero is a nearest feasible point. Hence this formula is a genuine projection, rather than an assumed update.

Use step $\eta = 1/2$, satisfying $0 < \eta < 2/L$, and initial point $x_0 = 1$. The genuine projected-gradient update is

$$T(x) = P_C(x - \eta f'(x)) = P_C(x/2).$$

For feasible $x \geq 0$ this equals $x/2$. Induction therefore gives

$$x_k = T^k(1) = 2^{-k} > 0 \quad \text{for every finite } k, \quad x_k \longrightarrow 0.$$

Define the actual active set as $I(x) = \{i : c_i(x) = 0\}$, with the single constraint indexed by $i = 1$. Then

$$I(x_k) = \emptyset \quad \text{for every } k, \quad I(x_*) = \{1\}.$$

Thus no finite iterate identifies the solution's active set, and in particular there is no finite N after which the active sets agree. This failure occurs despite genuine convergence to the unique solution, strong convexity, feasibility and a safe step.

The missing nondegeneracy is transparent: the optimal multiplier is zero because stationarity is $f'(0) - \lambda = 0$. Strict complementarity fails. Finite-identification results with appropriate nondegeneracy assumptions are not contradicted.

Formal verification. Lean verifies the actual real objective and gradient, convexity and closed epigraph, quadratic strong-convexity identity, closed convex feasible set, unique constrained minimizer, true metric-projection inequality, actual projected-gradient update and every iterate, convergence, and the indexed active sets with the quantified failure of finite identification. Lean 4.19.0 and Mathlib c44e0c8ee63ca166450922a373c7409c5d26b00b are pinned.